LegalEase

3. Project Design Phase

AI-Powered Legal Document Generator

Objective

Design the system architecture, module responsibilities, document flow, and output formatting.

System Architecture

- Frontend Layer: Streamlit handles user inputs, preview, editing and downloads.
- Backend Layer: FastAPI receives requests, validates data and routes generation logic.
- AI Layer: GeminiDocumentGenerator builds a structured prompt and calls Gemini generate_content.
- Formatting Layer: DOCX, PDF and TXT generation modules transform the generated text into downloadable documents.

Module Design

- main.py: initializes FastAPI, provides the root endpoint and includes API routes.
- routes.py: defines DocumentRequest and the /generate endpoint.
- ai_core/gemini_generator.py: integrates Gemini and generates the document text.
- Frontend app: collects inputs, displays previews, provides editing and download controls.
- Formatting utilities: sanitize text and generate DOCX, PDF and HTML preview.

UI Design

- Document type text input.
- Parties text area.
- Terms text area using semicolon-separated clauses.
- Effective date input.
- Generate Document button.
- Edit Document control.
- TXT, DOCX and PDF download options.
- Styled HTML preview.

Document Design

- DOCX: title, paragraphs, logo, Times New Roman formatting, terms table and footer.

- PDF: centered logo, section headings, bullet-style terms, header/footer branding.
- HTML: styled preview for viewing the generated legal document in the application.